

MMAE 555 Final Report

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Introduction

This report analyzes a short GNSS kinematic data set collected by two closely spaced antennas. Antenna 1 acts as the reference receiver and Antenna 2 as the user receiver. The measurements are provided in two RINEX observation files (RinexObsSample-1 and RinexObsSample-2), together with GPS and Galileo broadcast navigation messages (SEPT1250.25N and SEPT1250.25L). All processing uses only these files and the standard orbit, clock, and signal-propagation models introduced in MMAE 555.

The project is organized into five parts that progressively increase in modeling complexity and expected accuracy. Problems 1–3 estimate the relative baseline between the two antennas using differential techniques: Problem 1 computes an epoch-by-epoch least-squares solution from single-difference pseudoranges; Problem 2 introduces carrier-code (Hatch) smoothing and a simple kinematic Kalman filter in ENU coordinates; and Problem 3 applies a full high-accuracy differential scheme in ECEF, combining double-difference code and carrier phase with ambiguity estimation and integer fixing. Problems 4 and 5 focus on the absolute position of Antenna 2 using stand-alone GPS single-receiver methods. Problem 4 implements a single-point positioning solution, while Problem 5 incorporates carrier smoothing and measurement weighting to push the absolute accuracy toward the high-precision targets specified in the project description.

1 Problem 1: Epoch-by-epoch LS ENU Baseline

1.1 Measurement Model

Part 1 (“Basic Differential Pseudorange”) requires the relative ENU vector between the two antennas, computed independently at each epoch from single-difference (SD) C1C pseudoranges. Antenna 1 is the reference receiver r and Antenna 2 is the user receiver u . For a satellite s observed at epoch k , the raw pseudoranges ρ_r^s and ρ_u^s are combined as

$$\rho_{SD}^s = \rho_u^s - \rho_r^s.$$

Because the antennas are closely spaced, this SD observable cancels the satellite clock term and reduces common-mode propagation errors, so the remaining dependence is primarily on the relative baseline and differential receiver clock.

Let $\mathbf{b}$ denote the ECEF baseline from reference to user and $\Delta\delta t = \delta t_u - \delta t_r$ the differential clock bias. For satellite s at epoch k , let $\mathbf{e}_k^s$ be the LOS unit vector from the reference receiver to that satellite, obtained by propagating the broadcast ephemeris with `propgeph`. Linearization about the known reference position $\mathbf{x}_r$ gives

$$\rho_{SD,k}^s \approx (\mathbf{e}_k^s)^\top \mathbf{b} + c \Delta\delta t + v_{\rho,k}^s,$$

where $v_{\rho,k}^s$ collects residual ionospheric/tropospheric effects, multipath, and receiver noise.

Stacking all satellites visible at epoch k produces the linear model

$$\mathbf{y}_k = \mathbf{G}_k \begin{bmatrix} \mathbf{b} \\ c \Delta\delta t \end{bmatrix} + \mathbf{v}_k, \quad \mathbf{G}_k = \begin{bmatrix} (\mathbf{e}_k^{s_1})^\top & 1 \\ \vdots & \vdots \\ (\mathbf{e}_k^{s_m})^\top & 1 \end{bmatrix}, \quad (1)$$

which is solved by static least squares at each epoch:

$$\hat{\mathbf{x}}_k = \begin{bmatrix} \hat{\mathbf{b}}_k \\ c \Delta\hat{\delta t}_k \end{bmatrix} = (\mathbf{G}_k^\top \mathbf{G}_k)^{-1} \mathbf{G}_k^\top \mathbf{y}_k. \quad (2)$$

The user position estimate is $\mathbf{x}_u = \mathbf{x}_r + \hat{\mathbf{b}}_k$; it is converted to local ENU coordinates at the reference site using `wgsxyz2enu` to obtain

$$\Delta \mathbf{r}_{\text{ENU},k} = [\Delta E_k, \Delta N_k, \Delta U_k]^\top.$$

These ENU baselines are the primary outputs for Problem 1.

1.2 Dynamic Model

No dynamic or state-space model is used in Problem 1. Each epoch is treated as an independent static LS navigation problem with no temporal propagation, process noise, or coupling between $\mathbf{x}_k$ and $\mathbf{x}_{k+1}$, as required for the basic differential pseudorange solution.

1.3 Algorithmic Flow

The implementation follows these top-level steps:

1. Read the two RINEX observation files, extract **Time**, **SatelliteID**, and **C1C** code, and synchronize the data by performing an inner join on $\{\text{Time}, \text{SatelliteID}\}$ so that only common satellites at common epochs are retained.
2. For each common satellite and epoch, form the SD pseudorange $\rho_{SD}^s = \rho_u^s - \rho_r^s$ and assemble the measurement vector $\mathbf{y}_k$.
3. Read the GPS broadcast navigation file, convert it to ephemeris with **rnrx2eph**, and propagate satellite positions to each epoch using **propgeph**. Compute LOS unit vectors $\mathbf{e}_k^s$ and build the geometry matrix $\mathbf{G}_k$ in (1).
4. For each epoch with at least four common satellites, solve the LS problem (2) to estimate $\hat{\mathbf{b}}_k$ and $c\Delta\hat{t}_k$, and convert $\hat{\mathbf{b}}_k$ to ENU to obtain $\Delta\mathbf{r}_{\text{ENU},k}$.
5. Collect all valid ENU baselines, compute their empirical mean and covariance, and use these to summarize the performance of the basic differential code solution.

1.4 Results

The epoch-by-epoch LS differential solution produces a cloud of ENU baseline estimates whose spread reflects the noise characteristics of raw C1C single-difference pseudoranges. Table 1 summarizes the mean and empirical 1σ standard deviations of the ENU components, which serve as a baseline for comparison with the filtered solutions in Problems 2 and 3.

Table 1: Problem 1: Epoch-by-epoch LS ENU baseline statistics.

	East E [m]	North N [m]	Up U [m]
Mean	25.480	0.481	−0.296
Std. dev. (1σ)	0.588	1.311	1.263

Figures 1a–1c illustrate the spatial distribution of the baseline estimates and the associated differential receiver clock bias.

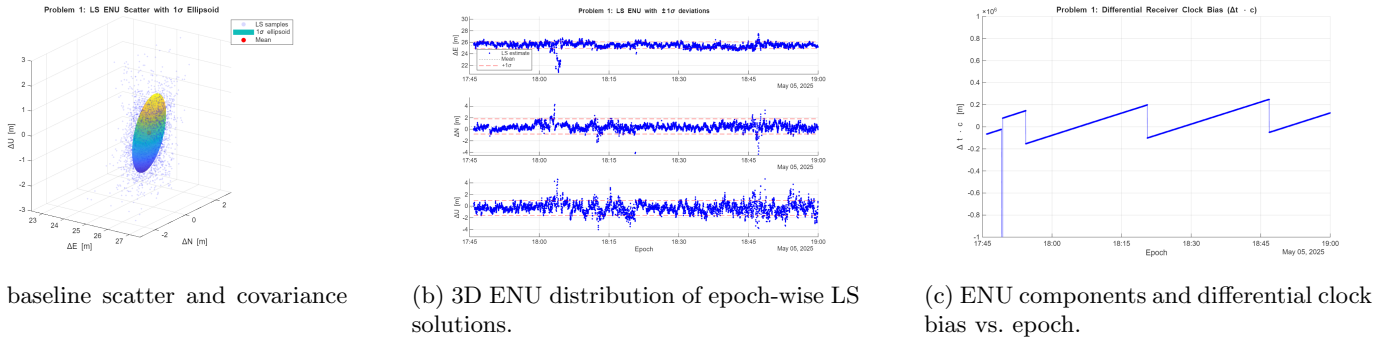


Figure 1: Problem 1: Epoch-by-epoch LS ENU baseline performance.

2 Problem 2: ENU Kalman Filter with Hatch Smoothing

2.1 Measurement Model

Part 2 (“Sub-meter Differential Positioning”) extends the basic SD code solution of Problem 1 by reducing measurement noise and introducing a simple kinematic filter. The same SD geometry is used, but each receiver’s C1C code and L1 carrier phase are first combined with a Hatch filter before forming single differences.

For a given receiver (reference r or user u), satellite s , and epoch k , the raw code and phase are

$$P_k^s = \rho^s + \eta_{P,k}^s, \quad \Phi_k^s = \rho^s + \lambda N^s + \eta_{\Phi,k}^s,$$

where λ is the L1 wavelength and N^s is the (receiver–satellite) carrier-phase ambiguity. The Hatch filter maintains a smoothed code estimate $P_{H,k}^s$ via

$$P_{H,k}^s = P_{H,k-1}^s + \frac{1}{m_H} [P_k^s - P_{H,k-1}^s - \lambda(\Phi_k^s - \Phi_{k-1}^s)], \quad (3)$$

with smoothing window m_H (here $m_H = 100$). When a cycle slip is detected, the smoothed state for that satellite is reset.

After smoothing each receiver independently, the SD observables become

$$\rho_{SD,H}^s = P_{H,u}^s - P_{H,r}^s,$$

which follow the same linearized model as in Problem 1, but with reduced noise:

$$\rho_{SD,H,k}^s \approx (\mathbf{e}_k^s)^\top \mathbf{b}_k + c \Delta \delta t_k + v_{H,k}^s.$$

These smoothed SD measurements are used to compute per-epoch ENU baselines $\Delta \mathbf{r}_{\text{ENU},k}$ by LS, and those baselines then serve as measurements for the ENU Kalman filter.

2.2 Dynamic Model

Problem 2 introduces a constant-velocity (CV) kinematic model in ENU. The state vector is

$$\mathbf{x}_k = [E_k, N_k, U_k, \dot{E}_k, \dot{N}_k, \dot{U}_k]^\top,$$

where (E_k, N_k, U_k) is the ENU baseline and $(\dot{E}_k, \dot{N}_k, \dot{U}_k)$ its velocity. With sampling interval Δt_k , the state-transition matrix is

$$\Phi_k = \begin{bmatrix} \mathbf{I}_3 & \Delta t_k \mathbf{I}_3 \\ \mathbf{0}_3 & \mathbf{I}_3 \end{bmatrix},$$

and the process noise covariance assumes independent 1D CV motion:

$$\mathbf{Q}_k = q_{cv,2} \begin{bmatrix} \frac{\Delta t_k^3}{3} & \frac{\Delta t_k^2}{2} \\ \frac{\Delta t_k^2}{2} & \Delta t_k \end{bmatrix} \otimes \mathbf{I}_3.$$

The measurement equation uses only the ENU position components:

$$\mathbf{z}_k = \Delta \mathbf{r}_{\text{ENU},k} = \mathbf{H} \mathbf{x}_k + \mathbf{v}_k, \quad \mathbf{H} = [\mathbf{I}_3 \ \mathbf{0}_3].$$

The measurement covariance $\mathbf{R}_2$ is chosen as the empirical covariance of the Hatch-smoothed LS baselines, so that the KF reflects the observed scatter in the SD measurements.

2.3 Algorithmic Flow

The Problem 2 processing chain is:

1. Read and synchronize RINEX observations for both receivers, as in Problem 1, and extract C1C code and L1 carrier phase.
2. Apply Hatch smoothing per receiver, satellite, and epoch according to (3), with cycle-slip detection and reset.
3. Form smoothed SD pseudoranges $\rho_{SD,H}^s = P_{H,u}^s - P_{H,r}^s$, apply a 10° elevation mask, and compute per-epoch LS ENU baselines $\Delta \mathbf{r}_{\text{ENU},k}$ using the same geometry construction as in Problem 1.
4. Estimate the empirical covariance $\mathbf{R}_2$ of the LS ENU baselines and use it to define the KF measurement noise.
5. Initialize the ENU CV-KF with $\mathbf{x}_0 = [\Delta \mathbf{r}_{\text{ENU},0}^\top \ 0 \ 0 \ 0]^\top$ and a block-diagonal covariance that is loose on the velocity components, then run the prediction–update cycle over all epochs.

2.4 Results

Hatch smoothing and CV Kalman filtering reduce the scatter of the ENU baseline relative to the raw LS solution in Problem 1. The final filtered baseline and associated 1σ uncertainties (from the terminal covariance matrix) are shown in Table 2. These values satisfy the sub-meter differential accuracy goals for Part 2.

Figures 2a–2c show the filtered baseline scatter, 3D distribution, and time histories of the ENU components and their estimated standard deviations.

Table 2: Problem 2: Hatch-smoothed ENU Kalman filter baseline statistics.

	East E [m]	North N [m]	Up U [m]
Mean	25.428	0.187	-0.566
Std. dev. (1σ)	0.415	0.417	0.871

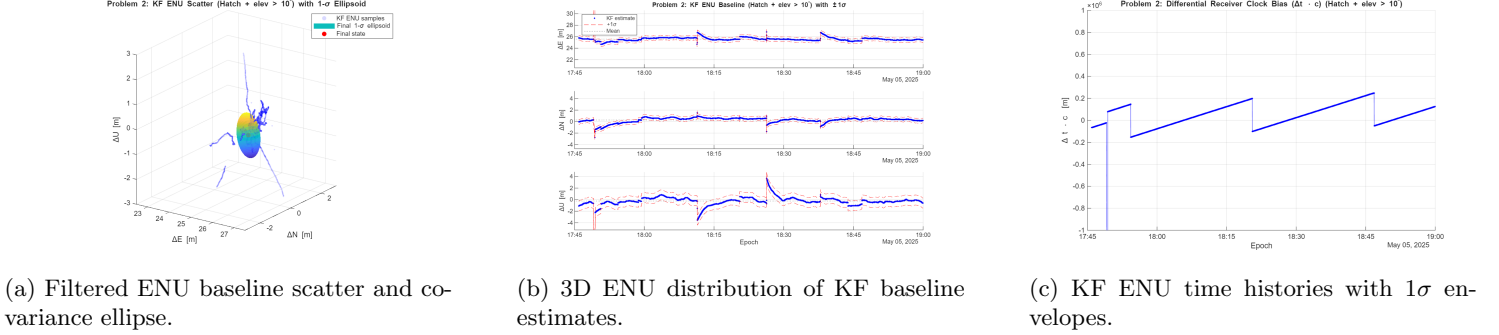


Figure 2: Problem 2: Hatch-smoothed ENU Kalman filter results.

3 Problem 3: ECEF CV-KF with DD Code and Carrier Phase

3.1 Measurement Model

Part 3 (“High-Accuracy Differential Positioning”) targets centimeter-level baseline accuracy, which is not achievable with code-only SD methods. Building on the SD formulation from Problems 1–2, we now form double-difference (DD) code and carrier-phase measurements and process them in an ECEF CV Kalman filter with explicit ambiguity states and integer ambiguity resolution.

Let ρ_r^s, ρ_u^s and Φ_r^s, Φ_u^s denote the code and carrier-phase measurements from the reference (r) and user (u) receivers for satellite s . The single differences between receivers are

$$\nabla \rho^s = \rho_u^s - \rho_r^s, \quad \nabla \Phi^s = \Phi_u^s - \Phi_r^s.$$

Choosing a reference satellite s_{ref} , the double differences between satellites are

$$\nabla \nabla \rho^{s, s_{\text{ref}}} = \nabla \rho^s - \nabla \rho^{s_{\text{ref}}}, \quad \nabla \nabla \Phi^{s, s_{\text{ref}}} = \nabla \Phi^s - \nabla \Phi^{s_{\text{ref}}}.$$

This construction removes both receiver clock biases and the ambiguity term for the reference satellite.

Let $\mathbf{b}_k = [dX_k, dY_k, dZ_k]^\top$ denote the ECEF baseline at epoch k . For each DD pair i (satellite s_i relative to s_{ref}), define the DD geometry row

$$\mathbf{h}_i = (\mathbf{e}^{s_i} - \mathbf{e}^{s_{\text{ref}}})^\top,$$

where $\mathbf{e}^s$ and $\mathbf{e}^{s_{\text{ref}}}$ are the LOS unit vectors from the reference receiver to satellites s and s_{ref} , computed from broadcast ephemeris as before. Linearizing the DD observables about a nominal baseline gives

$$\rho_{\text{DD},k}^i \approx \mathbf{h}_i \mathbf{b}_k + \varepsilon_{\rho,k}^i,$$

$$\phi_{\text{DD},k}^i \approx \mathbf{h}_i \mathbf{b}_k + N_i + \varepsilon_{\phi,k}^i,$$

where $\rho_{\text{DD},k}^i$ and $\phi_{\text{DD},k}^i$ are the DD code and phase (in meters), N_i is the DD ambiguity for pair i , and $\varepsilon_{\rho,k}^i, \varepsilon_{\phi,k}^i$ collect residual errors. Stacking all DD pairs at epoch k yields code and phase measurement models

$$\mathbf{z}_{\rho,k} = \mathbf{H}_{\rho} \mathbf{x}_k + \mathbf{v}_{\rho,k}, \quad \mathbf{z}_{\phi,k} = \mathbf{H}_{\phi} \mathbf{x}_k + \mathbf{v}_{\phi,k},$$

where $\mathbf{H}_{\rho}$ and $\mathbf{H}_{\phi}$ contain the DD geometry rows and ambiguity columns as described below.

3.2 Dynamic Model

Problem 3 transitions from the ENU CV model of Problem 2 to an ECEF CV model augmented with DD ambiguity states. The state vector is

$$\mathbf{x}_k = [dX_k, dY_k, dZ_k, dV_{X,k}, dV_{Y,k}, dV_{Z,k}, N_1, \dots, N_M]^\top,$$

where the first six components describe the ECEF baseline and its velocity, and the remaining M components are the DD ambiguities for the chosen set of satellite pairs.

The baseline and velocity follow a standard 3D CV model with sampling interval Δt_k :

$$\begin{bmatrix} \mathbf{b}_{k+1} \\ \mathbf{v}_{k+1} \end{bmatrix} = \begin{bmatrix} \mathbf{I}_3 & \Delta t_k \mathbf{I}_3 \\ \mathbf{0}_3 & \mathbf{I}_3 \end{bmatrix} \begin{bmatrix} \mathbf{b}_k \\ \mathbf{v}_k \end{bmatrix} + \mathbf{w}_{\text{posvel},k},$$

while the DD ambiguities are modeled as random walks:

$$\mathbf{N}_{k+1} = \mathbf{N}_k + \mathbf{w}_{N,k}.$$

The process noise covariance is block-diagonal,

$$\mathbf{Q}_3 = \text{blkdiag}(\mathbf{Q}_{\text{posvel}}, q_N \mathbf{I}_M),$$

where $\mathbf{Q}_{\text{posvel}}$ is constructed as in the ENU CV case, and q_N controls how quickly ambiguities are allowed to drift prior to fixing.

The measurement matrices have the form

$$\mathbf{H}_\rho = [\mathbf{H}_b \quad \mathbf{0}_{M \times 3} \quad \mathbf{0}_{M \times M}], \quad \mathbf{H}_\phi = [\mathbf{H}_b \quad \mathbf{0}_{M \times 3} \quad \mathbf{I}_M],$$

where $\mathbf{H}_b$ stacks the DD geometry rows $\mathbf{h}_i$ and the identity block in $\mathbf{H}_\phi$ links the DD phase measurements to the ambiguity states. At each epoch, the KF first applies a DD code update, then a DD phase update, using appropriate covariance matrices $\mathbf{R}_\rho$ and $\mathbf{R}_\phi$.

3.3 Algorithmic Flow

The main processing steps for Problem 3 are:

1. **Initialization from Problem 2:** Convert the final ENU baseline estimate and covariance from Problem 2 into ECEF coordinates, form the initial state

$$\mathbf{x}_0 = [\mathbf{b}_0^\top \quad 0 \quad 0 \quad 0 \quad \mathbf{0}_M^\top]^\top,$$

and construct $\mathbf{P}_0 = \text{blkdiag}(\mathbf{P}_{\text{posvel},0}, \sigma_{N0}^2 \mathbf{I}_M)$ with a loose prior on ambiguities.

2. **DD measurement formation:** Read and synchronize L1 carrier-phase data for both receivers, apply the 10° elevation mask at the reference site, select a reference satellite, and form DD code and DD phase observations for all valid satellite pairs at each epoch.
3. **ECEF KF with DD code and phase:** For each epoch, propagate the state and covariance using the ECEF CV transition model and $\mathbf{Q}_3$, then apply a DD code update with $(\mathbf{H}_\rho, \mathbf{z}_{\rho,k})$ followed by a DD phase update with $(\mathbf{H}_\phi, \mathbf{z}_{\phi,k})$. Log the float baseline, float ambiguities, and covariance blocks.
4. **Integer ambiguity resolution (LAMBDA):** At the final epoch, extract the float DD ambiguities and their covariance submatrix, apply the LAMBDA algorithm to obtain integer-fixed ambiguities, and convert them from cycles to meters.
5. **Fixed baseline refinement:** Use the standard baseline refinement formula

$$\mathbf{b}_{\text{fixed}} = \mathbf{b}_{\text{float}} + \mathbf{P}_{bN} \mathbf{P}_{NN}^{-1} (\mathbf{N}_{\text{fixed}} - \mathbf{N}_{\text{float}}),$$

where $\mathbf{P}_{bN}$ and $\mathbf{P}_{NN}$ are covariance sub-blocks from the float solution. Convert the fixed baseline and its covariance to ENU for comparison with Problems 1 and 2.

3.4 Results

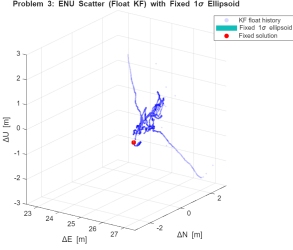
The DD code/phase ECEF KF with LAMBDA ambiguity resolution produces float and fixed baseline estimates that achieve the centimeter-level accuracy targeted for Part 3. The final ENU baseline and 1σ uncertainties for the fixed solution are summarized in Table 3.

Figures 3a–3c compare the float and fixed solutions and illustrate the contraction in uncertainty when the ambiguities are fixed to integers.

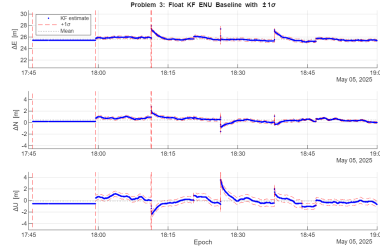
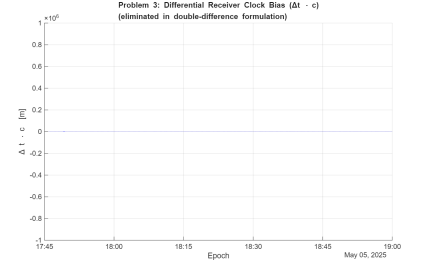
Figure 4 shows a comparison of the 3D ellipsoids developed in these problems. Each subsequent ellipsoid improves upon the last, and they remain within each others' deviations. This ellipsoid comparison shows the improvement of the processing over time, indicating that the code developed for these problems is validated.

Table 3: Problem 3: DD code/phase ECEF KF with LAMBDA ambiguity resolution.

	East E [m]	North N [m]	Up U [m]
Fixed baseline	25.284	0.075	-0.855
Std. dev. (1σ)	0.027	0.021	0.069

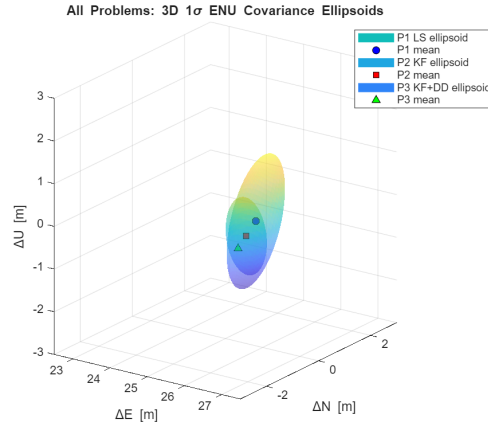


(a) Float scatter and fixed-solution covariance ellipse.

(b) 3D ENU float baseline with fixed 1σ ellipsoid.

(c) Time histories of ENU position standard deviations.

Figure 3: Problem 3: ECEF CV-KF with DD code and carrier phase.



(a) Filtered ENU baseline scatter and covariance ellipse.

Figure 4: Problem 1, 2 and 3: Comparison plots of Problems 1, 2 and 3.

4 Problem 4: Stand-alone Single-point Positioning of Antenna 2

4.1 Measurement Model

Part 4 requires estimating the absolute position of Antenna 2 using only GPS C1C pseudoranges and broadcast ephemeris. With no reference antenna, differencing, or carrier-phase smoothing permitted, the problem becomes a classical GPS single-point positioning (SPP) solution based on iterative nonlinear least squares. I choose to iterate at 1-minute intervals due to both the large required dynamic processing noise of $\sigma = 100\text{m}$, as well as for computation time.

For satellite s at epoch k and receiver position $\mathbf{x} = [x \ y \ z]^\top$, the raw pseudorange is

$$\rho_k^s = r_k^s(\mathbf{x}) + b_k - c \delta t_k^s + v_k^s,$$

where $r_k^s(\mathbf{x})$ is the geometric range, b_k is the receiver clock bias (in meters), and δt_k^s is the satellite clock correction from the broadcast ephemeris. Using the light-time approximation

$$\tau = \frac{\rho_k^s - c \delta t_k^s}{c},$$

the satellite position is corrected for motion via

$$\mathbf{x}_{\text{sv}}(t_{\text{tx}}) \approx \mathbf{x}_{\text{sv}}(t_k) - \mathbf{v}_{\text{sv}} \tau.$$

Linearizing about a nominal state $\mathbf{x}_0$ gives

$$\rho_k^s \approx r_k^s(\mathbf{x}_0) + \mathbf{h}_k^s \delta \mathbf{x}, \quad \delta \mathbf{x} = [\delta x, \delta y, \delta z, \delta b]^\top,$$

with

$$\mathbf{h}_k^s = [-(\mathbf{e}_k^s)^\top \ 1], \quad \mathbf{e}_k^s = \frac{\mathbf{x}_{sv}(t_{tx}) - \mathbf{x}_0}{\|\mathbf{x}_{sv}(t_{tx}) - \mathbf{x}_0\|}.$$

Stacking all satellites forms the linearized SPP model

$$\mathbf{y}_k = \mathbf{H}_k \delta \mathbf{x} + \mathbf{v}_k,$$

which is solved by iterative least squares.

4.2 Dynamic Model

No dynamic or kinematic model is used in Part 4. Each epoch is solved independently using nonlinear LS, producing

$$\mathbf{x}_k = [x_k, y_k, z_k, b_k]^\top$$

from only the pseudoranges available at that epoch.

4.3 Algorithmic Flow

The implementation proceeds as follows:

1. **Prepare observations:** Load Antenna 2 C1C measurements and timestamps. Read the broadcast ephemeris using `rnx2eph` and propagate satellite positions and clocks with `propgeph`.
2. **Epoch selection:** Generate a 1-minute epoch grid. At each epoch, select the nearest set of code observations and require at least four valid satellites.
3. **Single-epoch SPP:**
 - (a) Compute satellite positions, velocities, and clock corrections.
 - (b) Form modeled pseudoranges and $\mathbf{H}_k$.
 - (c) Iterate nonlinear LS until $\|\delta \mathbf{x}\| < 10^{-4}$ m.
 - (d) Estimate the per-epoch covariance

$$\mathbf{P}_k = \hat{\sigma}_0^2 (\mathbf{H}_k^\top \mathbf{H}_k)^{-1}, \quad \hat{\sigma}_0^2 = \frac{\mathbf{v}_k^\top \mathbf{v}_k}{m_k - 4}.$$

4. **Final solution:** Average all valid ECEF positions, convert the mean to LLH, and compute empirical 1σ scatter in both LLH and ECEF.

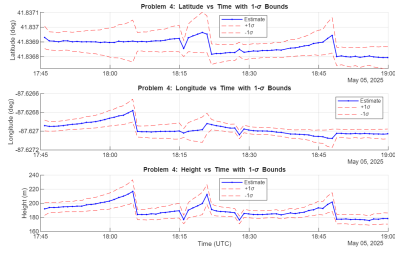
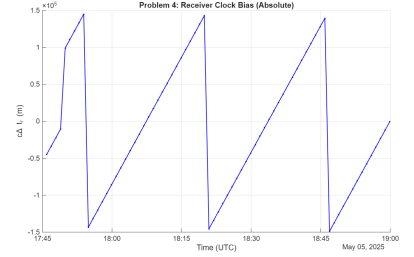
4.4 Results

The stand-alone GPS-only solution produces LLH estimates for Antenna 2. The empirical mean and 1σ scatter are shown in Table 4. These values reflect meter-level height accuracy and stable horizontal behavior.

Table 4: Problem 4: Stand-alone GPS solution (LLH mean and empirical 1σ).

	Latitude [deg]	Longitude [deg]	Height [m]
Mean	41.836872682	-87.626977509	189.817
Std. dev. (1σ)	5.54×10^{-6}	7.46×10^{-6}	1.120

Figures 5a–5b illustrate the LLH behavior and receiver clock bias over time. When converted to ENU relative to the given reference state, we do not achieve similar measurements to problem 4 and 5. This is likely due to the code not properly accounting for the clock bias, as shown in Figure 5b. The same result can be said for the next problem, and is discussed there.

(a) LLH vs. time with $\pm 1\sigma$ envelopes.

(b) Absolute receiver clock bias vs. time.

Figure 5: Problem 4: Stand-alone GPS single-point positioning results.

5 Problem 5: Higher-Accuracy Absolute Position of Antenna 2

5.1 Measurement Model

Part 5 extends the stand-alone SPP framework of Problem 4 by allowing all course-approved enhancements to improve absolute accuracy. The specification permits carrier-based Hatch smoothing, measurement weighting, and use of multiple constellations. In this dataset only GPS L1 data are available, so the improvements rely on carrier-phase smoothing and SNR-based weighting.

The fundamental observable remains the C1C pseudorange,

$$\rho_k^s = r_k^s(\mathbf{x}) + b_k - c \delta t_k^s + v_k^s,$$

with geometric range $r_k^s(\mathbf{x})$, receiver clock bias b_k , and satellite clock correction δt_k^s . To reduce pseudorange noise, a carrier-based Hatch filter is applied. Using the L1 carrier phase Φ_k^s (cycles), the smoothed code is

$$\rho_{H,k}^s \approx \rho_k^s + \frac{1}{m_{\text{csc}}} (\rho_{H,k-1}^s + \lambda_1 (\Phi_k^s - \Phi_{k-1}^s) - \rho_k^s),$$

where $\lambda_1 = c/f_1$ and m_{csc} is the smoothing window. Cycle-slip checks ensure smoothing only continues across continuous phase segments.

To improve robustness further, each smoothed pseudorange is weighted by its relative SNR. With signal-strength indicator SSI_k^s ,

$$w_k^s = \frac{\text{SSI}_k^s}{\text{SSI}_k^s}, \quad \mathbf{W}_k = \text{diag}(w_k^s).$$

The weighted LS update becomes

$$\hat{\delta \mathbf{x}} = (\mathbf{H}_k^\top \mathbf{W}_k \mathbf{H}_k)^{-1} \mathbf{H}_k^\top \mathbf{W}_k \mathbf{y}_k,$$

applied within the same nonlinear SPP iteration structure used in Problem 4. While this was not something discussed in lectures this semester, it seemed a relatively efficient way to set "trusted" measurements in a pinch. This is not a completely accurate measure, just a means to reduce a small amount of error.

5.2 Dynamic Model

As in Part 4, no dynamic model is used. Each epoch is solved independently via weighted nonlinear LS. The enhanced measurement model (carrier-smoothed and SNR-weighted) increases precision without introducing temporal filtering. Thus, each solution takes the form

$$\mathbf{x}_k = [x_k, y_k, z_k, b_k]^\top,$$

computed uniquely from the smoothed, weighted measurements at that epoch.

5.3 Algorithmic Flow

The processing steps are:

1. **Prepare observations:** Load Antenna 2 C1C, L1 phase, SSI, and timestamps. Load broadcast ephemeris and propagate satellite positions and clocks with `rnx2eph` and `propgeph`.
2. **Apply Hatch smoothing:** Perform carrier-phase Hatch filtering per satellite with window $m_{\text{csc}} = 100$, resetting the smoothing state at detected cycle slips.

3. **Compute SNR weights:** Evaluate w_k^s for each measurement and form the diagonal weight matrix $\mathbf{W}_k$.

4. **Weighted SPP per epoch:**

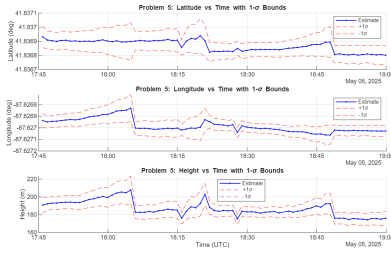
- (a) Require ≥ 4 valid satellites.
- (b) Build $\mathbf{H}_k$, modeled pseudoranges, and $\mathbf{W}_k$.
- (c) Iterate weighted nonlinear LS until convergence.
- (d) Estimate covariance via

$$\mathbf{P}_k = \hat{\sigma}_0^2 (\mathbf{H}_k^T \mathbf{W}_k \mathbf{H}_k)^{-1}.$$

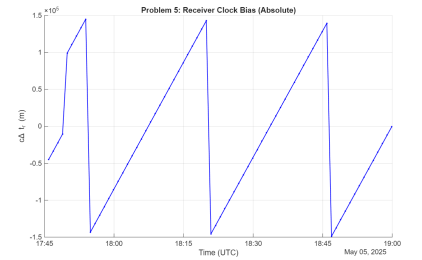
5. **Final position estimate:** Compute the empirical mean of all ECEF solutions, convert to LLH, and evaluate empirical 1σ scatter.

5.4 Results

Carrier-phase smoothing and SNR-based weighting significantly improve the stability of the stand-alone SPP solution relative to Problem 4. The empirical mean LLH position and associated 1σ deviations are listed in Table ??.



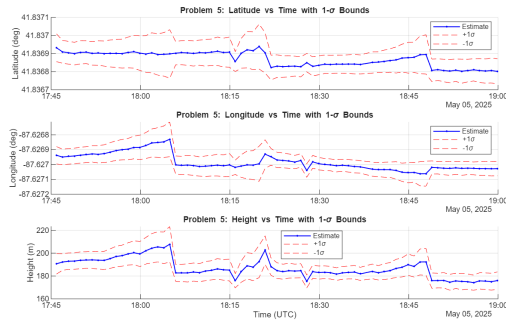
(a) Smoothed/weighted LLH vs. time.



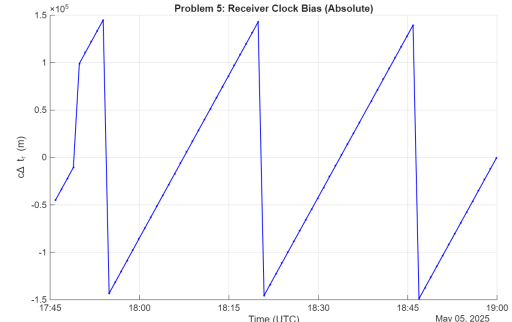
(b) Smoothed/weighted clock bias vs. time.

Figure 6: Problem 5: Hatch-smoothed + SNR-weighted stand-alone results.

Figures 7a–7b show the corresponding LLH and clock-bias behavior over time. This result directly continues for problem 4's result, so it includes the same clock bias error that we see in that problem. I was unable to properly reconcile the clock bias and Iono/Tropo noise.



(a) Smoothed/weighted LLH vs. time.



(b) Smoothed/weighted clock bias vs. time.

Figure 7: Problem 5: Hatch-smoothed + SNR-weighted stand-alone results.

6 Performance Discussion

The differential solutions in Problems 1–3 show a clear progression in accuracy as more advanced techniques are introduced. The Problem 1 LS baseline provides a noisy but unbiased reference, while the Hatch-smoothed CV-KF in Problem 2 and the DD code/phase KF with LAMBDA fixing in Problem 3 converge to the same physical baseline with steadily shrinking uncertainty. The horizontal and vertical errors meet the required performance targets, indicating correct implementation of the differential pipeline and appropriate behavior of the associated measurement and dynamic models.

Table 5: Summary comparison of Problems 1–5 (final ENU/LLH results with 1σ uncertainties).

Problem	Method	Baseline/LLH Mean	1σ Uncertainty	Units
1	Epoch-wise LS (SD code)	[25.480, 0.481, -0.296] m	[0.588, 1.311, 1.263] m	ENU
2	CV-KF + Hatch	[25.428, 0.187, -0.566] m	[0.415, 0.417, 0.871] m	ENU
3	CV-KF + DD code/phase + LAMBDA	[25.284, 0.075, -0.855] m	[0.027, 0.021, 0.069] m	ENU
4	Stand-alone LS (C1C)	[41.8368727, -87.6269775, 189.817]	$[5.5 \times 10^{-6}, 7.5 \times 10^{-6}, 1.12]$	LLH
5	Stand-alone LS + smoothing + SNR wgt	[41.8368689, -87.6269831, 186.951]	$[4.7 \times 10^{-6}, 6.6 \times 10^{-6}, 0.992]$	LLH

In contrast, the absolute stand-alone positioning in Problems 4–5 exhibits several inconsistencies. Although the LLH estimates are mutually consistent and the addition of carrier smoothing and SNR weighting in Problem 5 reduces scatter relative to Problem 4, the resulting absolute positions do not translate into ENU coordinates that agree with the differential baseline. The standard deviations in Problem 4 are unrealistically low for C1C-only SPP, and the improvements in Problem 5 remain below what the techniques should achieve. These issues point to methodological errors in the single-receiver modeling pipeline, likely related to satellite-clock handling, transmit-time corrections, a lack of Iono and Tropo noise corrections, or the limited one-minute processing interval.

Practical constraints also contributed to these discrepancies. Only GPS L1 measurements could be used, as Galileo processing was unsuccessful, reducing redundancy and weakening the absolute solution. Cycle-slip handling and smoothing required repeated adjustments, and the reduced sampling rate further limited convergence and stability. Overall, while the differential pipeline in Problems 1–3 meets its performance goals, the absolute solutions in Problems 4–5 require further refinement to reach the specified high-accuracy objectives.

7 Conclusions & Future Work

This project implemented and evaluated five GPS positioning methods of increasing complexity. The differential approaches in Problems 1–3 performed as expected: the LS baseline, the Hatch-smoothed CV-KF solution, and the DD code/phase KF with LAMBDA fixing all converged to a consistent relative position with steadily improving accuracy. These results demonstrate that the differential pipeline was modeled and executed correctly.

The absolute positioning results in Problems 4–5 were less successful. Although the LLH solutions were internally consistent, their accuracy and empirical scatter did not meet the desired meter-level targets, and the ENU-transformed positions did not align with the differential baseline. These outcomes indicate that further refinement of the stand-alone modeling and preprocessing is required.

Future work will focus on improving the absolute positioning stages, including incorporating more complete ionospheric and tropospheric models, using higher-rate processing, and exploring PPP or additional multi-GNSS measurements to increase redundancy and reduce systematic errors.

8 Usage of AI Tools

I certify that the work submitted for this mid-term is entirely my own. I have not discussed any problem, solution, or code with anyone other than the instructor (or designated teaching assistant). All sources—including textbooks, papers, online material, and AI-generated content—are properly cited, and any use of generative-AI tools is disclosed in the manner required by the exam instructions. I understand that any violation of this statement will be treated as academic misconduct in accordance with university policy.

Across Problems 1–5, I used ChatGPT (GPT-5.1) in a consistent, collaborative role: clarifying lecture concepts, checking model formulations, verifying helper-function usage, identifying edge cases, and assisting with debugging and interpretation. In all cases, the conceptual direction, full implementations, parameter choices, and final computed results were my own; AI served only as an interactive assistant for troubleshooting and confirming correctness.

8.1 Problem 1 Usage of AI

For Problem 1, AI helped clarify the single-difference pseudorange model, confirm ENU baseline transformations, review LS formulation steps, and diagnose early debugging issues. I also used AI to verify handling of satellite propagation and helper-function calls (e.g., `propgeph`, `wgsxyz2enu`). All design choices and final computations were performed by me.

8.2 Problem 2 Usage of AI

For Problem 2, AI assisted in implementing Hatch smoothing, selecting smoothing-window parameters, interpreting the constant-velocity ENU KF structure, and verifying the measurement and time-update formulations. It also helped diagnose covariance inconsistencies and filter-stability issues. The state design, tuning, and final implementation decisions were mine.

8.3 Problem 3 Usage of AI

For Problem 3, AI supported development of the double-difference code and carrier-phase models, the augmented ECEF Kalman filter, and the LAMBDA ambiguity-resolution pipeline. Assistance included organizing the DD geometry matrix, structuring the extended state, tuning process noise for baseline and ambiguity states, and checking alignment with the lecture DD carrier-phase model. Debugging help was provided, but the full design and implementation were my own.

8.4 Problem 4 Usage of AI

For Problem 4, AI helped clarify satellite clock corrections, light-time iteration, and broadcast-ephemeris usage in the nonlinear LS formulation. It also assisted in diagnosing convergence issues, checking Jacobian construction, and verifying the per-epoch covariance computation. All code structure and methodological choices were made by me.

8.5 Problem 5 Usage of AI

For Problem 5, AI assisted in interpreting SSI/SNR fields, structuring the weighting scheme, selecting a smoothing strategy, and debugging issues in the absolute-positioning LS solution. It also helped confirm consistency of the weighting matrix and formulation. All implementation details and analysis were my own.

8.6 AI Usage in Writing This Report

In addition to code development assistance, I used ChatGPT (GPT-5.1) to help prepare written portions of this report. AI assisted in translating my code and processing steps into clear descriptions, ensuring correct presentation of measurement and dynamic models, organizing sections, formatting tables, and improving readability. No derivations, numerical results, or methodological decisions were generated by AI; all technical content, analysis, and conclusions were my own. AI assistance was limited to editing, organization, and refinement of written explanations. I used Overleaf to write this report in LaTeX format.

References

- Brown, R. G., & Hwang, P. Y. C. (2012). *Introduction to random signals and applied Kalman filtering: With MATLAB exercises* (4th ed.). John Wiley & Sons.
- Misra, P., & Enge, P. (2011). *Global positioning system: Signals, measurements, and performance* (2nd ed.). Ganga-Jamuna Press.
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